# Dhana Karthikeya Ventrapragada

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## Education

**University of California, Berkeley** **Expected Dec 2027**

*Bachelor of Arts in Data Science and Applied Mathematics | GPA: 3.85/4.00* *Berkeley, CA*

- **Relevant Coursework:** Principles & Techniques of Data Science, Probability for Data Science, Introduction to Urban Data Analytics; *In progress:* Introduction to Machine Learning, Efficient Algorithms & Intractable Problems

## Technical Skills

**Languages**: Python, SQL, Java

**ML & Statistics**: scikit-learn, XGBoost, LightGBM, PyTorch, Cross-Validation, Feature Engineering

**Data**: pandas, NumPy, GeoPandas, Matplotlib, seaborn, PostgreSQL, ArcGIS Online, Jupyter

**Engineering**: FastAPI, Claude API, Apify, REST APIs, AWS S3, pytest, Git, GitHub

## Experience

**IDX Exchange | Data Science Intern | Remote** **Jun 2026 - Aug 2026**

- Developed a machine learning model that predicts California home sale prices, typically within **8.6%** of the actual price, using **71K** single-family sales cleaned from **143K** raw records.
- Benchmarked four model types and selected a tuned XGBoost ($R^2$ of **0.88**), finding that deeper trees improved accuracy more than a lower learning rate.
- Resolved a data leakage bug that had invalidated early results, and nearly halved a simpler model's error (from **29%** to **15%**) by engineering a school district feature.
- Analyzed errors by price band to find weak spots: luxury homes (**13%** error vs. **6.5%** mid-priced) and low-sales towns.

**Guang Labs | Engineering Intern | Remote** **May 2026 - Aug 2026**

- Engineered a service for an AI video startup that classifies videos and images by type (sports, gaming, talking) using Gemini, retrying uncertain or malformed responses on different clips, up to **10** attempts.
- Persisted each label with the model's confidence and reasoning so the team can audit it, backed by **86** tests.
- Developed an in-app notification bell that alerts users when their video finishes processing or fails.
- Built the service on Pydantic schemas and a deterministic fake provider, so it can be tested without live services.

## Projects

**Hindsight (Cal Hacks 13.0)** | *Claude, FastAPI, React, Apify, Factory Droid* GitHub

- Co-designed a transfer planner for California community college students that shows which classes count toward their major using official ASSIST data, citing a source for every answer.
- Shaped the product idea, features, and vision, and gave feedback that improved the interface and features.
- Kept it grounded in real data: credits are matched by code against official records, and campus opportunities are scraped from real pages (Apify, run by a Droid agent), with source links.

**Housing Price & Assessment Bias** | *Python, pandas, scikit-learn* GitHub

- Built a model that estimates home values from **200K+** Cook County (Chicago) property records, and audited its fairness across price ranges to show which homes get over- or under-valued and taxed unfairly.
- Engineered **15+** features (log square footage, polynomial age and location terms) and validated with 4-fold cross-validation (test RMSE **0.562**).

**SB 79 Transit Housing Analysis (Los Angeles)** | *ArcGIS Online, StoryMaps* GitHub

- Mapped the parts of Los Angeles that qualify for denser housing near transit under a new California law (**6,413** eligible zoning areas), layering in displacement risk, pollution, and fire hazard data in an interactive StoryMap.
- Wrote the Methods, Results, and Discussion of the research paper and policy brief that accompany the StoryMap (two-person project).

## Leadership & Involvement

**UC Berkeley CDSS | Undergraduate Course Staff, CDSS 198 | Berkeley, CA** **Aug 2026 - Present**

- Run exam reviews and write practice worksheets for CS and data science classes for transfer students, and give resume feedback.

**UC Berkeley | Undergraduate Course Staff, CS 61A | Berkeley, CA** **Jun 2026 - Aug 2026**

- Taught **4** weekly exam-prep sections in Python, Scheme, and SQL, reinforcing data structures like linked lists, trees, lists, and dictionaries, held office hours, and helped write midterm and final questions.